

BITS, PILANI – K. K. BIRLA GOA CAMPUS

Theory of Computation

(CS F51)

Lecture No. 10



Main Theorem

Two finite automata M_1 & M_2 are *equivalent* if and only if $L(M_1) = L(M_2)$

It means the two automata's generate the same language may be *using different methods*.

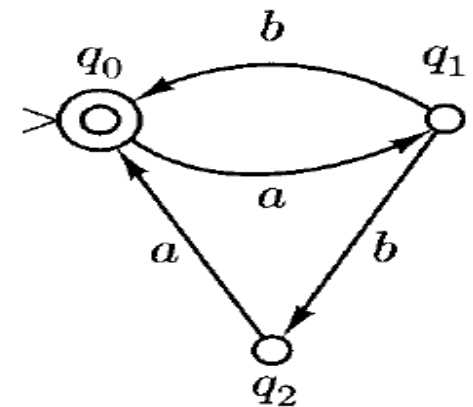
Theorem:

For each NFA, there is an equivalent DFA.

Proof: Key Idea

For pair (q_1, b) *think* of NFA as occupying *set of states* $\{q_0, q_2\}$

rather than a single undetermined member of that set.



The *construction* formalizes this idea.

Proof Continued

Let $M = (K, \Sigma, \Delta, s, F)$ be a *NFA*.

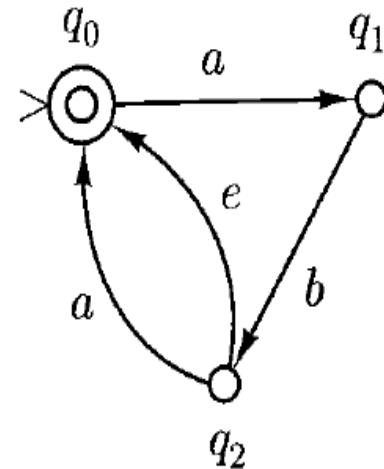
To construct $M' = (K', \Sigma, \delta, s', F')$ *DFA* such that
 $L(M) = L(M')$

The *set of states of M'* will be $K' = 2^K$ all subsets of K

The *set of final states of M'* will consists of all those subsets of K that contains *at least one final state* of M .

Transition function for M'

A *move* of M' on reading $a \in \Sigma$ *imitates* a move of M on input symbol a & *possibly* followed by any number of *e-moves* of M .



Proof Continued

Definition:

For any state $q \in K$, $E(q)$ denote the set of all states of M that are *reachable from state q* without reading any input.

So $E(q) = \{p \in K : (q, e) \vdash_M^* (p, e)\}$

Formal definition of DFA $M' = (K', \Sigma, \delta, s', F')$

$$K' = 2^K$$

$$s' = E(s)$$

Easy to see that M' is an DFA

$$F' = \{Q \subseteq K : Q \cap F \neq \emptyset\}$$

Remains to prove that
 $L(M) = L(M')$

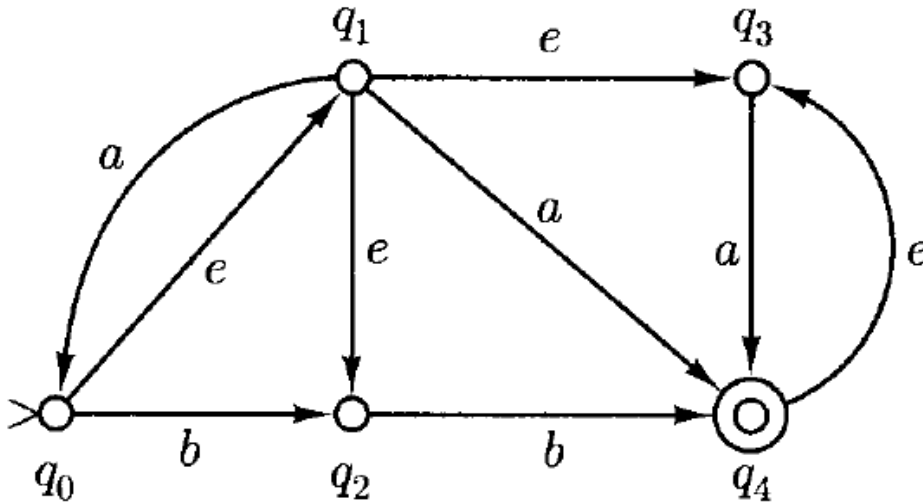
For each $Q \subseteq K$ & each $a \in \Sigma$, define

$$\delta(Q, a) = \cup \{E(p) : p \in K \text{ \& } (q, a, p) \in \Delta \text{ for some } q \in Q\}$$

$\delta(Q, a)$ is set of all states on M to which M can go by *reading* input a & possibly following *several e transitions*.

Proof Continued

Example

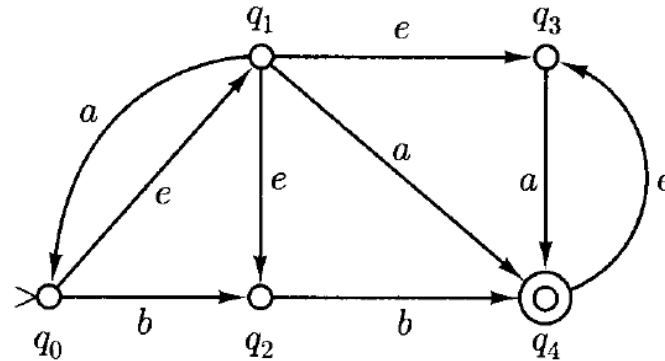


Then $s' = E(s) = \{q_0, q_1, q_2, q_3\}$

NFA to DFA Construction

Example

Consider NFA M



Note

Since M has 5 states, corresponding DFA M' will have $2^5 = 32$ states.

However, in practice only a few of these states will be *reachable* from initial state s' by reading some input string. We shall build this *reachable* part of M' by starting from s' & introducing a new state only when it is needed.

NFA to DFA Construction

$$s' = E(q_0)$$

$$\delta(Q, a) = \cup \{E(p) : p \in K \text{ \& } (q, a, p) \in \Delta \text{ for some } q \in Q\}$$

$$= \{q_0, q_1, q_2, q_3\}$$

$$(q_1, a, q_0), (q_1, a, q_4)$$

$$\& (q_3, a, q_4)$$

Are all the transitions

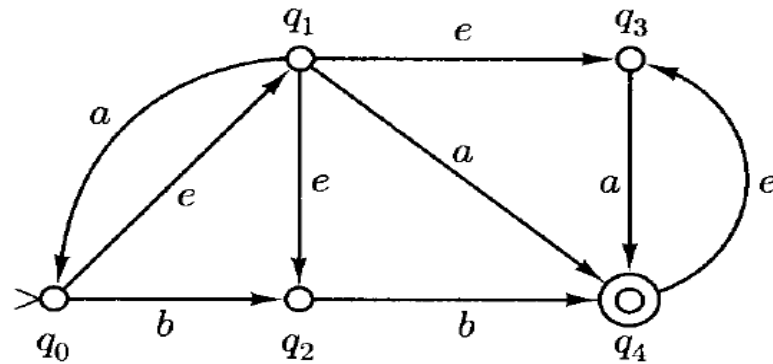
(q, a, p) for some $q \in s'$

$$\therefore \delta(s', a) = E(q_0) \cup E(q_4) = \{q_0, q_1, q_2, q_3, q_4\}$$

Similarly, $(q_0, b, q_2), (q_2, a, q_4)$

Are all the transitions (q, b, p) for some $q \in s'$

$$\therefore \delta(s', b) = E(q_2) \cup E(q_4) = \{q_2, q_3, q_4\}$$



NFA to DFA Construction

The *new states* introduced

Are $\{q_0, q_1, q_2, q_3, q_4\}$ and
 $\{q_2, q_3, q_4\}$

We *repeat* the construction
the newly introduced states

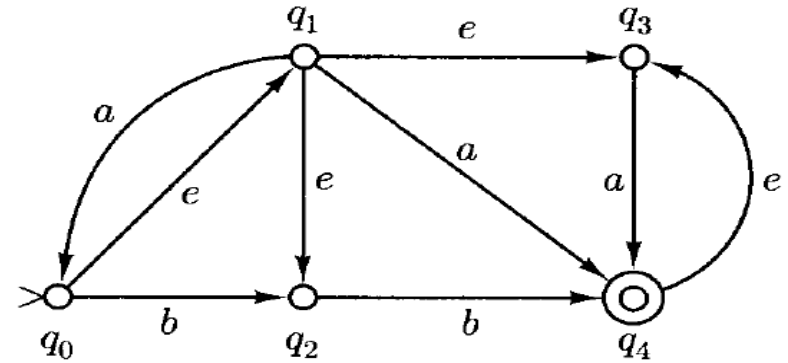
$$\delta(\{q_0, q_1, q_2, q_3, q_4\}, a) = E(q_0) \cup E(q_4) = \{q_0, q_1, q_2, q_3, q_4\}$$

$$\delta(\{q_0, q_1, q_2, q_3, q_4\}, b) = E(q_2) \cup E(q_4) = \{q_2, q_3, q_4\}$$

$$\delta(\{q_2, q_3, q_4\}, a) = E(q_4) = \{q_3, q_4\}$$

$$\delta(\{q_2, q_3, q_4\}, b) = E(q_4) = \{q_3, q_4\}$$

The *new state* introduced is $\{q_3, q_4\}$



NFA to DFA Construction

Next

$$\delta(\{q_3, q_4\}, a) = E(q_4) = \{q_3, q_4\}$$

$$\delta(\{q_3, q_4\}, b) = \phi$$

The *new state* introduced is ϕ

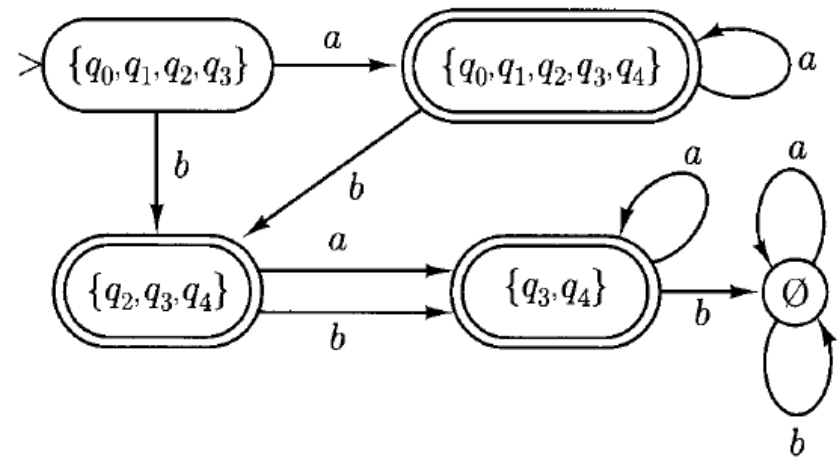
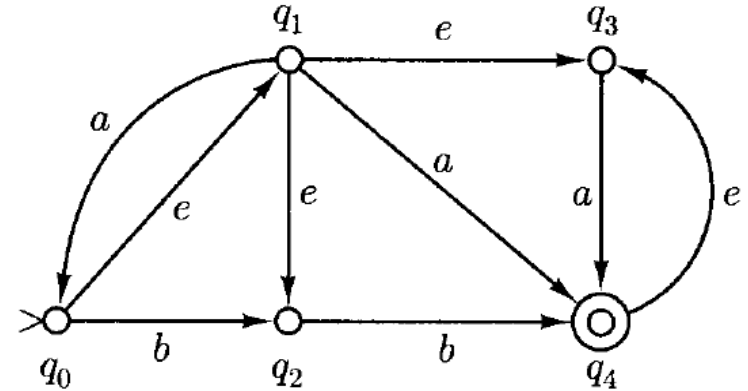
And finally,

$$\delta(\phi, a) = \delta(\phi, b) = \phi$$

So *no new state* is discovered & we *terminate* the process.

F' the set of *final states*

contains each set of states of which q_4 is a member.



Proof Continued

Claim:

For any string $w \in \Sigma^*$ & any $p, q \in K$

$(q, w) \vdash_M^* (p, e)$ if and only if $(E(q), w) \vdash_{M'}^* (P, e)$ for some set P containing p

We prove the claim by *induction on $|w|$*

Base Case: $|w| = 0$ or $w = e$

We must show $(q, e) \vdash_M^* (p, e)$ if and only if $(E(q), e) \vdash_{M'}^* (P, e)$ for some set P containing p

Note, $(q, e) \vdash_M^* (p, e)$ is $\Leftrightarrow p \in E(q)$

Since M' is an DFA $(E(q), e) \vdash_{M'}^* (P, e) \Leftrightarrow P = E(q)$

& $p \in P$ i.e $p \in E(q)$

This proves the *base case*

Proof Continued

Inductive Hypothesis: Suppose the claim is true for all strings of length $\leq k$, for some $k \geq 0$

Inductive Step: Let w be string of length $k+1$

Let $w = va$ where $a \in \Sigma$ & $v \in \Sigma^*$ with $|v| = k$

Suppose $(q, w) \vdash_M^* (p, e)$.

Then there are states r_1 & r_2

such that $(q, w) \vdash_M^* (r_1, a) \vdash_M (r_2, e) \vdash_M^* (p, e)$.

Now $(q, va) \vdash_M^* (r_1, a) \Leftrightarrow (q, v) \vdash_M^* (r_1, e)$ where $|v| = k$

$\therefore$ by **inductive hypothesis** $(E(q), v) \vdash_{M'}^* (R_1, e)$ for some set R_1 containing r_1

Since $(r_1, a) \vdash_M (r_2, e) \Rightarrow (r_1, a, r_2) \in \Delta$

$\therefore$ by construction of M' , $E(r_2) \subseteq \delta(R_1, a)$

Also $(r_2, e) \vdash_M^* (p, e) \Rightarrow p \in E(r_2)$ & $\therefore p \in \delta(R_1, a)$

$\therefore (R_1, a) \vdash_{M'} (P, e)$ for some set P containing p

Thus $(E(q), va) \vdash_{M'}^* (R_1, a) \vdash_{M'} (P, e)$

We leave the **converse** part as homework

Proof Continued

Proof of the Theorem follows easily from the Claim.

Let $w \in \Sigma^*$

$w \in L(M)$

$\Leftrightarrow (s, w) \vdash_M^* (f, e)$ for some $f \in F$

$\Leftrightarrow (E(s), w) \vdash_{M'}^* (Q, e)$ for some Q containing f (By Claim)

$\Leftrightarrow (s', w) \vdash_{M'}^* (Q, e)$ for some $Q \in F'$

$\Leftrightarrow w \in L(M')$

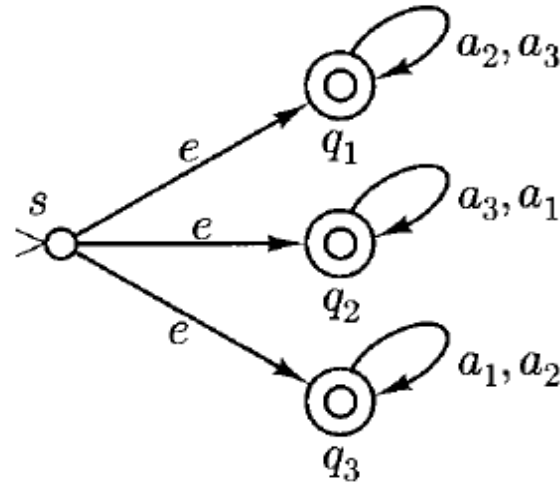
NFA to DFA Construction

Example: Let $\Sigma = \{a_1, a_2, a_3\}$

$L = \{w \in \Sigma^* : \text{there is a symbol } a_i \in \Sigma \text{ *not* appearing in } w\}$

NFA accepting L is

The *construction* of the DFA in this case is *exponential*.



Construction is left as **Exercise**.

FSA & Regular Languages

Theorem:

The class of languages accepted by FSA is *closed* under

1. Union
2. Concatenation
3. Kleene Star
4. Complementation
5. Intersection

Once we prove 1 & 4, **5** follows from *De Morgan's Law*

We have *already seen* results 1 & 4 for DFA

Note:

We have proved similar result for *regular languages*

FSA & Regular Languages

Intuitive idea

We have seen this Example before

Example: Consider the NFA M

This NFA *accepts* the language

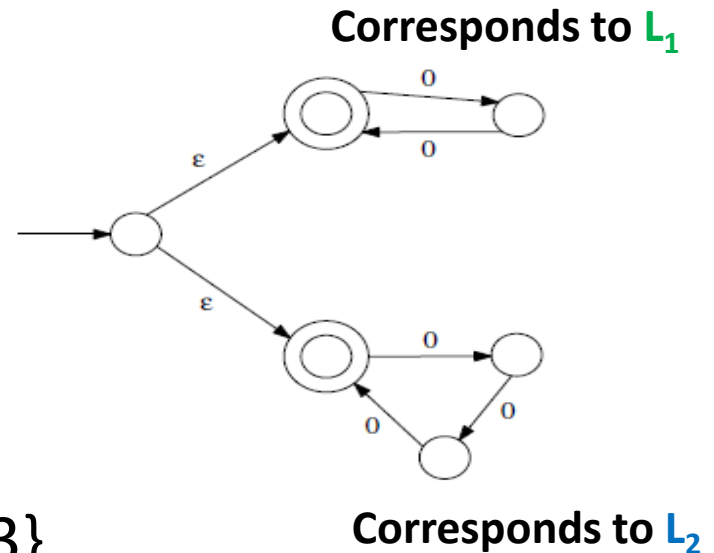
$$L = \{0^k : k \equiv 0 \pmod{2} \text{ or } k \equiv 0 \pmod{3}\}$$

Observe that L is the **union** of the two languages

$$L_1 = \{0^k : k \equiv 0 \pmod{2}\} \text{ \& } L_2 = \{0^k : k \equiv 0 \pmod{3}\}$$

Note

The NFA M is *a new initial state* and *empty transitions* to initial states of the FSA corresponding to L_1 & L_2



FSA & Regular Languages

Proof (for *union*):

Suppose L_1 & L_2 are languages with $L_1 = L(M_1)$ & $L_2 = L(M_2)$ where

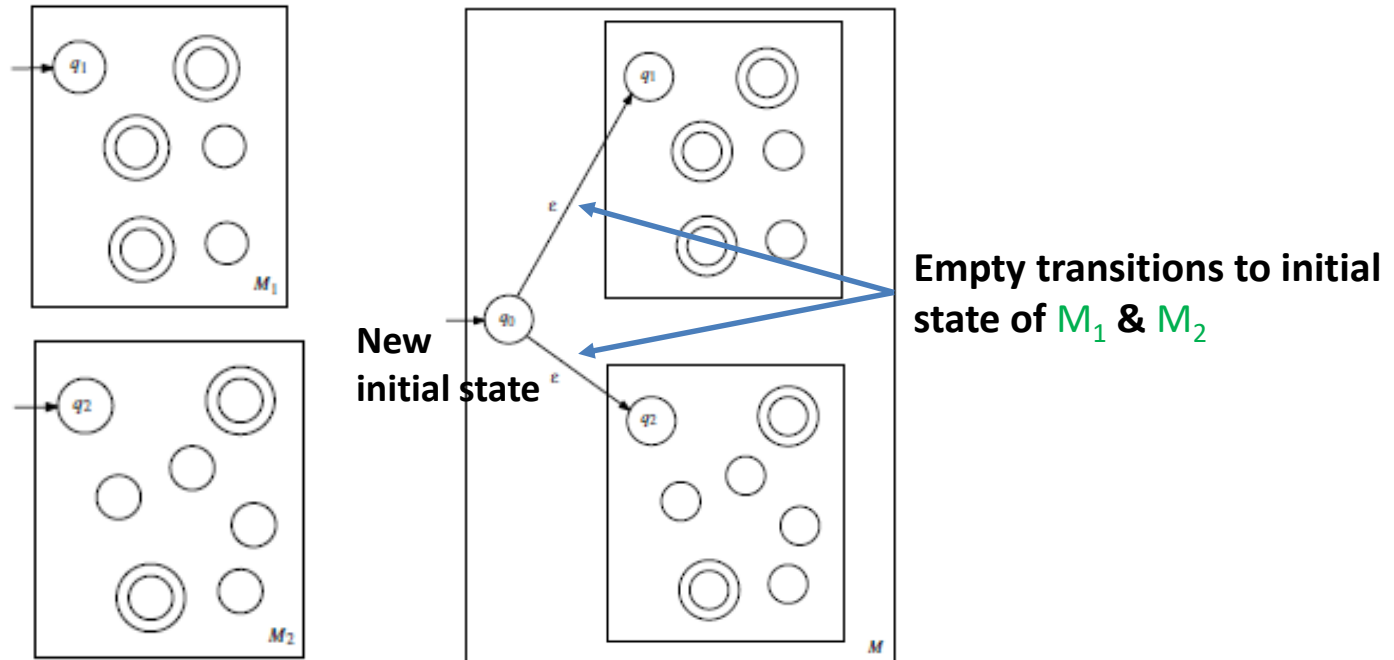
$M_1 = (K_1, \Sigma, \Delta_1, q_1, F_1)$ & Let $M_2 = (K_2, \Sigma, \Delta_2, q_2, F_2)$

We assume $K_1 \cap K_2 = \emptyset$, otherwise *rename* the states

We will construct $M = (K, \Sigma, \Delta, q_0, F)$ such that $L(M) = L(M_1) \cup L(M_2)$

We *imitate* the idea of the example:

M begins any computation by *nondeterministically* choosing to enter initial state of M_1 or M_2 & then *imitates* either M_1 or M_2



FSA & Regular Languages

Formally

The NFA M is defined as follows:

$M = (K, \Sigma, \Delta, q_0, F)$, where q_0 is a new state not in K_1 or K_2

$$K = K_1 \cup K_2 \cup \{q_0\}$$

$$F = F_1 \cup F_2$$

$$\Delta = \Delta_1 \cup \Delta_2 \cup \{(q_0, e, q_1), (q_0, e, q_2)\}$$

$$w \in L(M) \Leftrightarrow (q_0, w) \vdash_M^* (q, e) \text{ for some } q \in F$$

$$\Leftrightarrow \text{either } (q_1, w) \vdash_M^* (q, e) \text{ for some } q \in F_1$$

$$\text{or } (q_2, w) \vdash_M^* (q, e) \text{ for some } q \in F_2$$

$$\Leftrightarrow w \in L(M_1) \cup L(M_2)$$

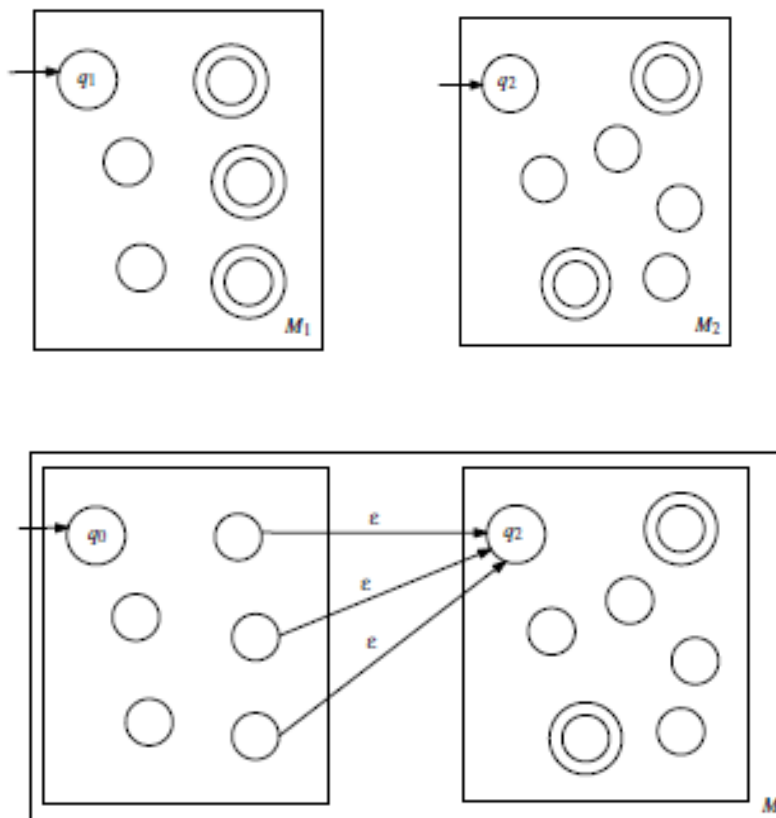
FSA & Regular Languages

For concatenation

We will construct $M = (K, \Sigma, \Delta, q_0, F)$ such that $L(M) = L(M_1) L(M_2)$

Intuitive idea

Start by *simulating* M_1 for a while & then *moving nondeterministically* (*guessing* the part from L_1) from a *final state* of M_1 to the *initial state* of M_2 & then *imitate* M_2



FSA & Regular Languages

Formally

The NFA M is defined as follows:

$$M = (K, \Sigma, \Delta, q_0, F)$$

$$K = K_1 \cup K_2$$

$$q_0 = q_1 \quad F = F_2$$

$$\Delta = \Delta_1 \cup \Delta_2 \cup \{(f, e, q_2) : f \in F_1\}$$

$$w \in L(M) \Leftrightarrow (q_0, w) \vdash_M^* (q, e) \text{ for some } q \in F$$

$$\Leftrightarrow (q_1, w) \vdash_M^* (q, e) \text{ for some } q \in F_2$$

$$\Leftrightarrow (q_1, w) \vdash_M^* (q_1, \alpha\beta) \vdash_{M_1}^* (f_1, \beta) \vdash_M (q_2, \beta) \vdash_{M_2}^* (f, e)$$

$$\text{for some } \alpha \in L(M_1) \ \& \ \beta \in L(M_2) \ \& \ f_1 \in F_1 \ \& \ f \in F_2$$

$$\Leftrightarrow w \in L(M_1) L(M_2)$$